Jacob Chase Palko

704-794-2586 | jpalko4@charlotte.edu | www.linkedin.com/in/jacob-c-palko

EDUCATION

University of North Carolina at Charlotte, Charlotte, NC
B.S. Mechanical Engineering, Expected Graduation - May 2027
Relevant Coursework: Engineering Design, Circuit Analysis, Manufacturing Processes; **ENGINEERING PROJECT EXPERIENCE**
Automated drone | ongoing

- Designed and built a heavy-lift drone with semi-automated flight capabilities, establishing a scalable hardware and software foundation to support fully autonomous payload delivery in the future.

Air Engine Design & Fabrication | Ongoing

- Manufactured a fully functional air engine, ensuring precise component fitment with strict **.001** inch tolerances.
- Machining parts on manual lathes and mills

Automated Dorm Room Door Opener | Spring 2025

- Engineered an automated door opening mechanism capable of pulling **8 pounds** of force by designing a motorized spool system integrated with custom circuitry and power systems.
- Programming an Arduino-based control system in Python and conducting iterative testing, ultimately delivering an intuitive push-button user interface.

Automated Pick-and-Place Robot | Team Lead | Fall 2026

- Leading a team in developing a four-axis robotic arm with a magnetic gripper to autonomously locate, pick up, and sort steel spheres.
- Designed and analyzed the RA3 elbow joint, performing torque, stress, deflection, and safety factor calculations to guide servo selection and evaluate structural performance.

LEADERSHIP & INVOLVEMENT

- **UNCC Helping Hands** – Prosthetics Team Member
- **Get Out The Vote Initiative** – Team Leader, Ardrey Kell High School.
- **Eagle Scout** – Boy Scouts of America, Troop Leader

WORK EXPERIENCE

- **Lowe's Home Improvement -** Warehouse | Summer 2025
- **Regal Cinemas** – Shift Lead | May 2021 – August 2024

CERTIFICATIONS/SKILLS

- **Certificates:** AutoCAD and Microsoft Word
- **Computer Skills**: Proficient in AutoCAD and Creo, MS Office (Excel, Word, PowerPoint), and Google Workspace, Arduino and microcontroller programming.
- **Other skills**: Machining, 3D Printing